

Automated Flower Classification



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Project Details

Research Question – How accurately can a CNN-based feature extraction classify flower images based on features such as petal shape and color?

Motivation – Recognizing and protecting diverse flower species plays a crucial role in helping our ecosystem thrive.

Modeling Approach – Classify flower species by training a VGG16 Convolutional Neural Network (CNN) model on 8,000+ images of 102 different flower classification types.

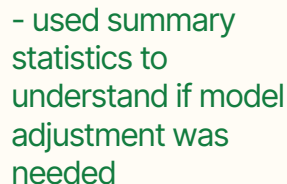
Data Acquisition

- Downloaded 3 datasets from Oxford
 1. JPG folder of 8,000+ flower images
 2. Image ID and index
 3. Index and flower name
- The image dataset contains 102 flower classes, and each class consists of 40-258 images
- Kept the background for environmental context
- Merged the data into one final dataset → 'Final_Dataset.csv'

Filename	Name of JPG file
Path	File path
Image	Flower image, expressed as pixel values
Image_id	A unique numerical value corresponding to each individual image
Index	A numerical value corresponding to the flower classification
Name	The name/classification of the flower itself, which corresponds to the index



- Increased epochs from 10 to 20 after early signs of underfitting



```
Epoch 8/10
205/205 [=====] - 229s 1s/step - loss: 3.0215 - accuracy: 0.2126 - val_loss: 2.2872 - val_accuracy: 0.4524
Epoch 9/10
205/205 [=====] - 228s 1s/step - loss: 2.9552 - accuracy: 0.2212 - val_loss: 2.0914 - val_accuracy: 0.4988
Epoch 10/10
205/205 [=====] - 228s 1s/step - loss: 2.8980 - accuracy: 0.2339 - val_loss: 2.0169 - val_accuracy: 0.5049
```

Tricky Analysis Decision

Choosing Number of Epochs:

- Too few → underfitting; too many → risk of overfitting and longer queue times on Rivanna.
- Settled on 20 epochs after observing continued improvement in validation accuracy beyond 10 epochs.

Resource Constraints:

- GPU queue delays on Rivanna made testing multiple models/epochs difficult.
- Prioritized training depth over model experimentation.

Model Architecture Selection:

- Considered simpler CNNs and other model types
- Chose VGG16 for strong performance on image classification tasks with limited tuning.



Bias and Uncertainty

- Unequal distribution of flower types
 - There are significantly more passions flowers and petunias
 - The model might perform worse on underrepresented flower types
- Since we opted to include the background, potential obstructions and/or variation in image quality may impact the models' ability to identify key features of flower classes
- The CNN model could potentially focus on unimportant features such as background, lighting etc.

Classification Performance

- Confusion matrix shows strong performance on the top 20 most frequent flower types.
- Some confusion remains between visually similar classes (e.g., classes 43, 95, and 83).

[illegible]

Next Steps

- Train model on more than 102 flower types/classes
 - Will not only improve the model's accuracy, but also increase its potential for real-world application
- Add more layers for higher accuracy → model will learn more complex patterns, which will help distinguish between similar flower classes
- Try a different model such as ResNet50

References

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**THANK
YOU!**